



UPSC Prelims

Science & Tech

Class No. 7

Health and Nanotechnology

Tuberculosis



- Recently, World Health Organization (WHO) released new guidance to support National Strategic Planning (NSP) for the tuberculosis (TB).
- NSP for TB : guides how to comprehensively address TB epidemic through goals and strategies.
- Specific, Measurable, Attainable, Realistic, Time bound criteria.
- Update to 2015 document.

India's efforts



- National Strategic Plan 2017 – 2025 : Eliminate TB by 2025.
- NTEP : Earlier this program called RNTCP.
- Nikshay Portal : National TB information system, National data repository
- Nikshay Poshan Yojana : Provides DBT to all TB patients towards nutritional support, by MoHFW, financed by GoI and World Bank.
- TB mukt bharat : PM's initiative by 2025
- Patho detect kit : Indigenous TB kit Indian lab

Other Efforts



- Moscow Declaration, 2017.
- WHO end TB strategy : incidence reduction by 80%, TB deaths by 90%, eliminate catastrophic costs.
- Find all, Treat all, End TB : Partnership between WHO, Stop TB partnership and Global Fund to end TB.
- BPaL : 6 month oral regimen for TB using 3 drugs, Bedaquilline, Pretomanid, Linezolid
- Reduced time of treatment : 18 months to 6 months. Made by TB alliance, a Not for profit organisation.



Tuberculosis

TB is caused by bacteria (*Mycobacterium tuberculosis*) and it most often affects the lungs

Types of TB



Pulmonary TB
(affects lungs)



Extra Pulmonary TB
(affects both lungs and other organs)



For prevention of TB

BCG (Bacille Calmette-Guérin) vaccine is used among children at the time of birth



Drug resistant TB



Multi- Drug Resistance TB (MDR- TB)

Does **not** respond to at least **isoniazid and rifampicin** (2 of the most powerful anti-TB drugs)



Extensively Drug Resistant TB (XDR- TB)

Does **not** respond to at least four of the core anti-TB drugs.



Totally drug-resistant tuberculosis (TDR-TB)

TB which is resistant to all the anti- TB drugs.

Fats

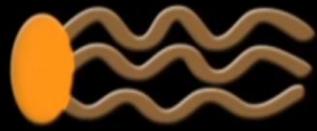


- We have to understand :
- Transfats
- Saturated vs Unsaturated
- Good Lipoproteins vs Bad Lipoproteins.

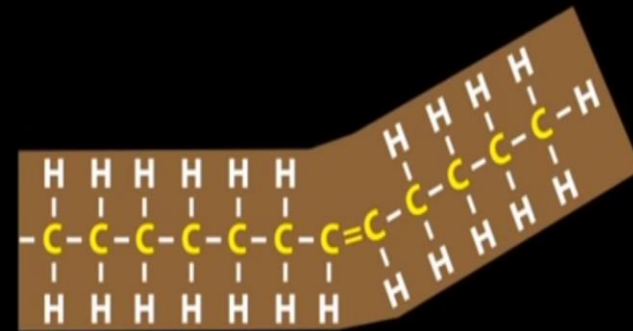
Fats



Saturated Fats	Unsaturated	Transfats
Solid @ room temperature	Liquid @ room temperature	Solid @ room temperature.
Even Natural are bad for you	Plant oils : good for you	artificially formed by hydrogenation of unsaturated fats. Some natural sources.
No double bonds	Have double bonds	Have very few double bonds, very bad for you
Animal Origins	MUFA, PUFA Omega 3 fatty acids, Omega 6 fatty acids.	Partially hydrogenated oils. This is done to increase the shelf life.



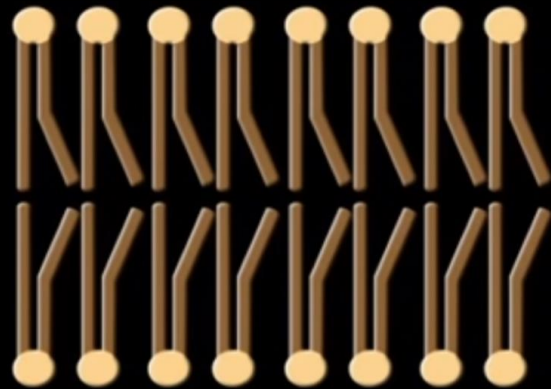
Saturated



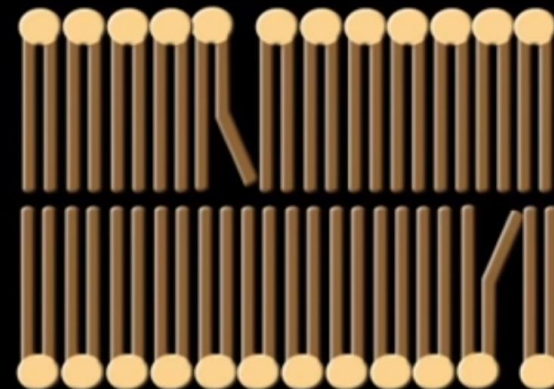
Unsaturated



Optimal unsaturated/saturated ratio

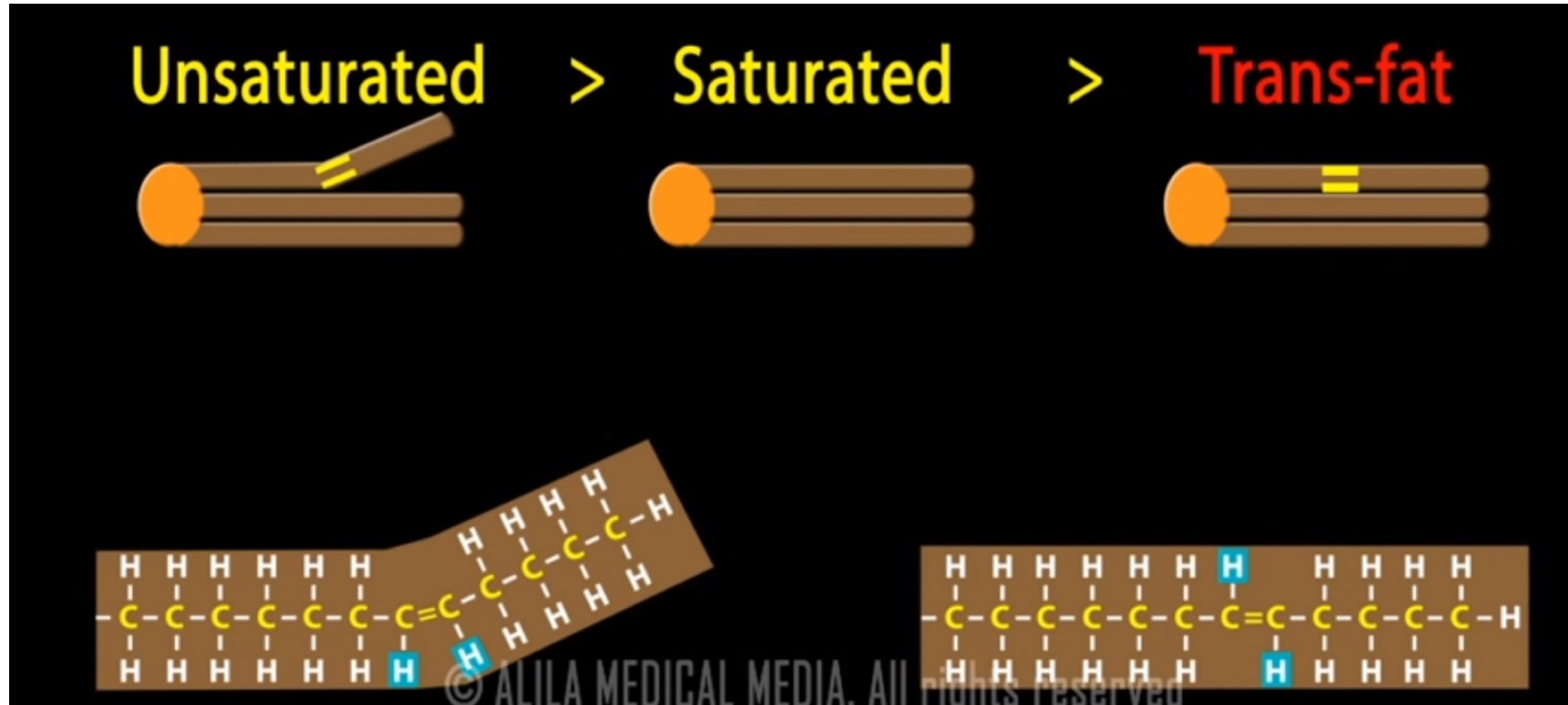


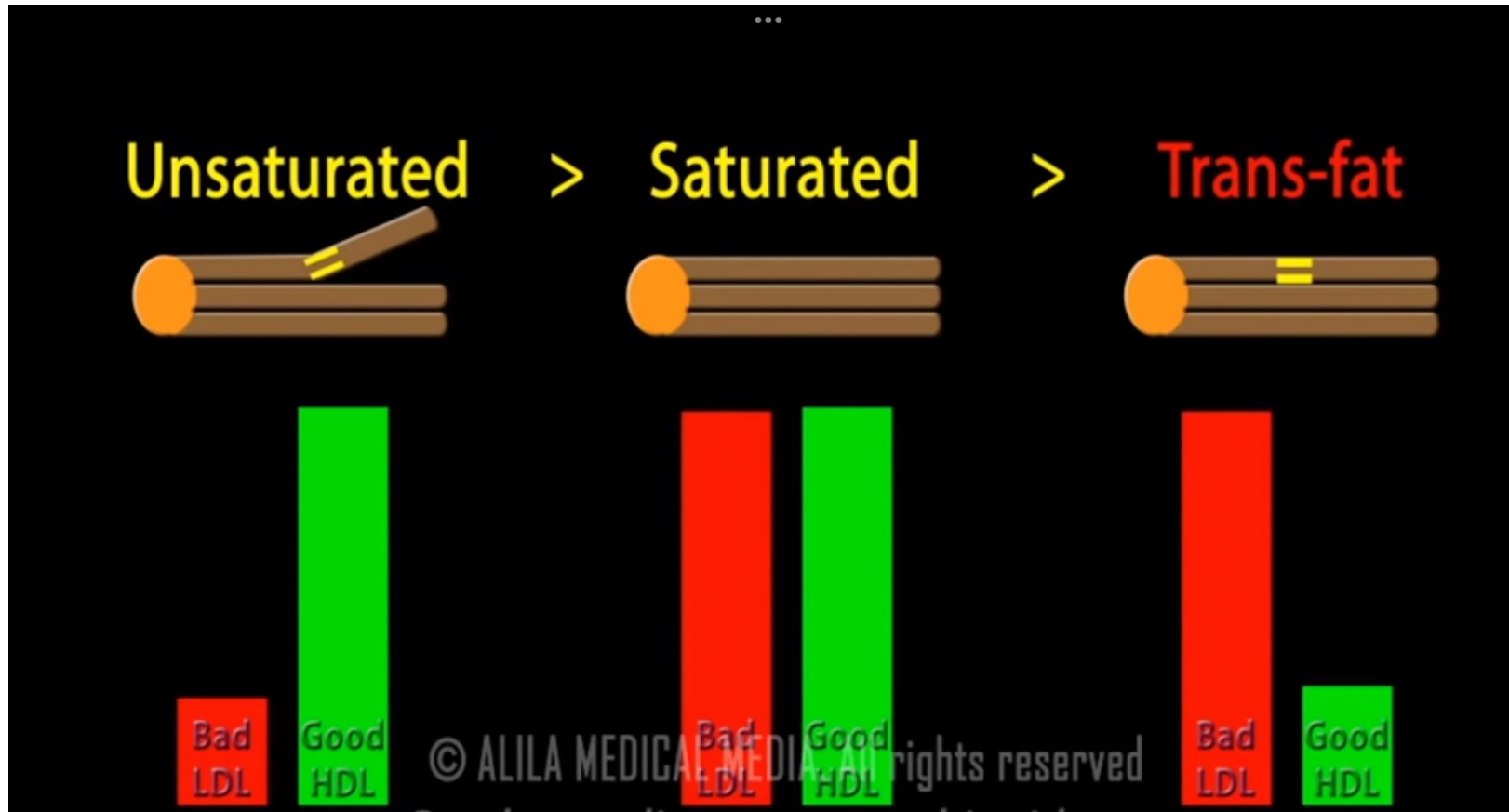
Too much saturated fat



Unsaturated > Saturated







Trans-Fats

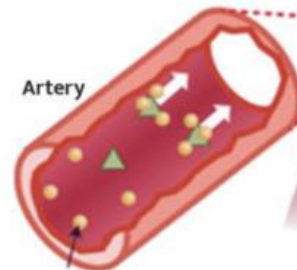


- Transfat, or trans-fattyacids, are unsaturated fatty acids that come from either natural or industrial sources:
- Naturally occurring trans-fat come from ruminants (cows and sheep).
- Industrially produced trans-fat are found in hardened vegetable fats such as margarine and are often present in snack foods, baked goods and fried foods.
- Industrially produced trans-fats are formed in a process that adds hydrogen to vegetable oil converting the liquid into a solid, resulting in “partially hydrogenated” oil (PHO).
- Manufacturers often use them as they have a longer shelf life and are cheaper than other fats.
- Health Impacts: Intake of TFA is associated with increased risk of heart attacks and death from coronary heart disease.
- Approximately 540,000 deaths each year globally and 60,000 in India can be attributed to intake of industrially produced TFA.

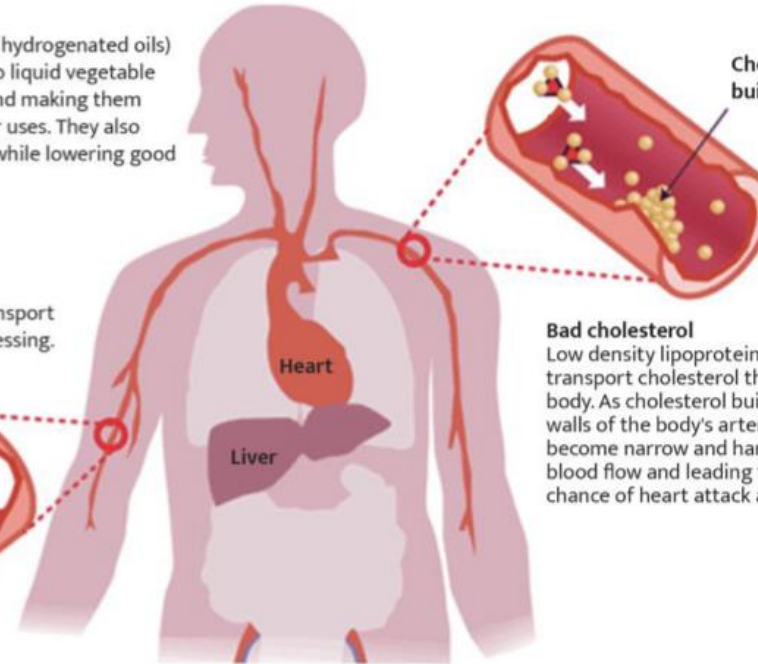
What they do to the body

Trans fats (also known as partially hydrogenated oils) are created by adding hydrogen to liquid vegetable oil, giving them longer shelf life and making them more suitable for frying and other uses. They also raise bad (LDL) cholesterol levels while lowering good (HDL cholesterol).

Good cholesterol
High-density lipoproteins (HDL) pick up excess cholesterol and transport it back to the body's liver for processing.



Cholesterol

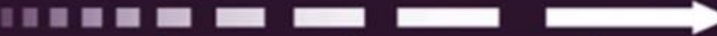


Cholesterol build-up

Bad cholesterol
Low density lipoproteins (LDL) transport cholesterol throughout the body. As cholesterol builds up in the walls of the body's arteries, the arteries become narrow and hardened, reducing blood flow and leading to an increased chance of heart attack and stroke.

Where it comes from

Trans fats are created by hydrogenation, a chemical process that hardens vegetable oil. The more solid the fat, the more trans fat it contains.

Low trans fat 				
				
Vegetable oil	Semi-liquid margarine	Soft margarine	Stick margarine	Shortening

Initiatives around trans fats



- WHO : trans fat elimination report called Countdown to 2023.
- Replace action framework : to eliminate industrially produced TFA from food.
- WHO recommendation : Transfat intake should be less than 1% of energy uptake.
- Part of Triple billion target of WHO :
 - 1 billion more people benefitting from universal health coverage
 - 1 billion more people better protected from health emergencies
 - 1 billion more people enjoying better health and well-being.

Indian Initiatives



- Target to make country TFA Free
- Phased target : reduce TFA in all food products containing all oils and fats less than 2% from 1 January 2022.
- Trans fat free logo : Food Contains less than 0.2g per 100g or 100ml and if edible oils or fats contain less than 1g of TFA per 100ml or 100g.
- Multilingual Mass media campaign called heart attack rewind campaign.

Foodborne Diseases



- South-East Asia Region contributes a quarter of the global burden of FBD morbidity and nearly 42% of FBD mortality
- FBD costing India almost 15 billion USD annually

Classification of Foodborne Diseases



- Food-borne infections – caused by consuming foods or liquids contaminated with bacteria, viruses, or parasites.
- These pathogens cause infection by invading and multiplying in the lining of the intestines and/or other tissues
- Food-borne intoxications - caused by consuming foods or beverages contaminated with a toxin.
Sources of toxins: Certain Bacteria, Poisonous chemicals, Natural toxins found in animals, plants, and fungi

Steps taken by FSSAI for Food Safety



- Eat right movement
- Blissful Hygienic Offering to God (BHOG)
- Hygiene Rating Scheme
- Food Safety Compliance System (FoSCoS)

Antioxidants



- Regular intake of fruits and vegetables neutralizes the free radical produced in the body during metabolism

State Food Safety Index



- Released by FSSAI annually: no need to study components

**Food Safety and Standards
Authority of India (FSSAI)**


New Delhi, India

**About:** Established, under administration **Ministry of Health & Family Welfare**, for laying down science based standards for articles of food to ensure safe food for human consumption.

**Concerned Law:** Food Safety and Standards Act, 2006.

**Mandate:**

- ▶ Laying down mechanisms and guidelines for accreditation of certification bodies.
- ▶ Provide scientific advice and technical support to Central Government and State Governments.
- ▶ Collect data regarding food consumption, incidence and prevalence of biological risk, identification of emerging risks and introduction of rapid alert system.
- ▶ Contribute to the development of international technical standards for food, sanitary and phyto-sanitary standards.



Food Fortification



- Consumer Affairs Ministry had issued uniform specifications for fortified rice kernels (FRK) for Grade A and common rice for procurement of fortified rice stocks, wherein 1% of FRK should be blended with normal rice
- Fortified rice: distributed under public distribution system and midday meals
- FSSAI norms: fortified rice contain iron, folic acid and Vitamin B-12. Rice may also be fortified with micronutrients such as Zinc, Vit A, Vit B1, Vit B2, Vit B3 and Vit B6

About Fortification



- Regulation: Under Food Safety Standards (Fortification of Foods) Regulation, 2018.
- Deliberately increasing the content of an essential micronutrient, i.e vitamins and minerals.
- Successful for iodized salt: Iodine-deficient countries has decreased from 54 to 32 since 2003.
- In India, wheat flour and rice are fortified with Iron, Vit B12 and folic acid, Milk and Edible Oil with Vit A and D and Double Fortified Salt with Iodine and Iron.

Standard Operating Procedure



- Issued by Department of food and public distribution to maintain desired quality standards for implementation of **rice fortification**.
- **Rice Specification:** Milled rice in which Fortified Rice Kernels (FRK) will be blended should comply with rice specification as per regulation.
- **Packaging of fortified food:** Fortified food package shall contain: **Fortified with (name of the fortificant)** and **+F logo** on the label.

About Standard Operating Procedure - Cont'd

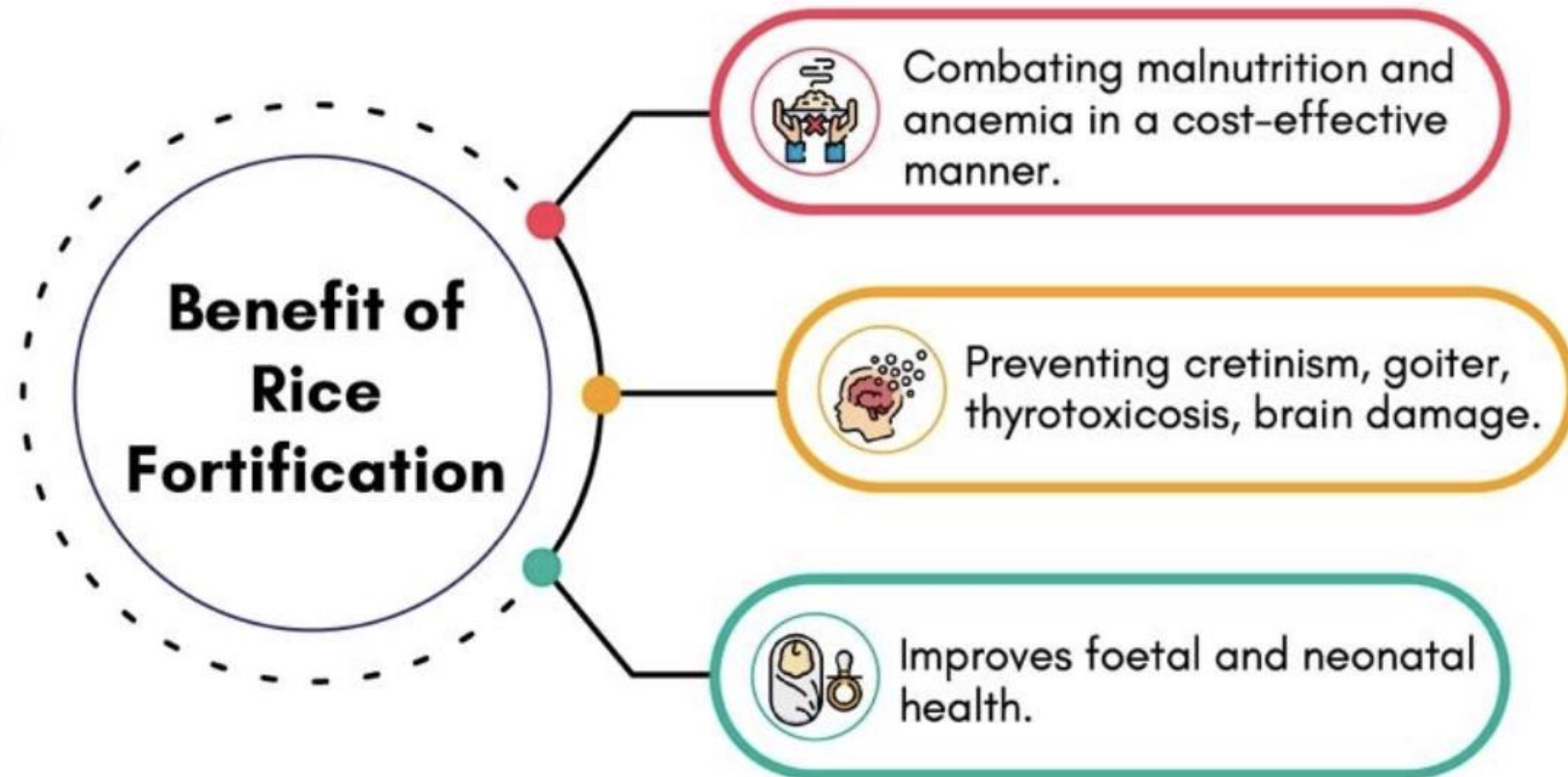


- May also carry a tagline Sampurna Poshan Swasth Jeevan
- Food fortified with Iron shall carry a warning statement for People with Thalassemia
- Rice when fortified shall contain mandatory micronutrients (Iron, Folic acid and Vit B12), or optional (Zinc, Vit A, Thiamine, Riboflavin, Niacin and Pyridoxine as per levels given by Food safety and Standards (Fortification of Foods) Regulations 2018

Technologies to add micronutrients



- Dusting: Adds micronutrients onto the surface of the rice grains using electrostatic force. Provides limited nutrient protection when rice is washed, soaked or cooked in excess water
- Coating and extrusion: Two step process. coating or extrusion technology is used to produce fortified kernels and the fortified kernels are blended with non-fortified rice at a ratio of 0.5% to 2%.
- In India, rice is fortified using extrusion technology in which milled rice is pulverized and mixed with a premix. FRK is added to traditional rice in ratio ranging from 1:50 to 1: 200



Forms of food fortification



- **Mass fortification:** micronutrients to foods commonly consumed by the general public
- **Targeted fortification:** specific subgroups of the population
- **Market-driven fortification:** business-oriented initiative.

Benefits of food fortification



- Addressing Hidden hunger
- Safe method
- Does not require any changes in food habits of people
- Does not alter the characteristics of the food—the taste, the feel, the look.

Steps towards fortification in India



- Food fortification in India began in the 1950s with vegetable oil fortification and salt iodization
- 10th, 11th ,12th Five Year Plans, POSHAN Abhiyan (National Nutrition Mission) and Anaemia-Mukt Bharat Mission
- In 2016, FSSAI imposed regulations on fortification for staples
- Other Steps taken by FSSAI include:
 - A dedicated unit called the Food Fortification Resource Centre
 - capacity building trainings and developed Information Education & Communication (IEC) material specifically for POSHAN Maah
 - Eat Right India movement

Oral Rehydration Solution



- Dr Dilip Mahalanabis who pioneered the proper practical, emergency use of oral rehydration solution, for diarrhoeal diseases passed away recently.
- Dr Dilip Mahalanabis was working in refugee camps during the 1971 Bangladesh Liberation war
- ORS: mixture of electrolytes (salts) and carbohydrates (in the form of sugar) dissolved in water. Used to replace salts and water that the body loses during dehydration caused by gastroenteritis, diarrhoea or vomiting. Contains potassium and sodium.
- These components maximize fluid absorption in the gastrointestinal tract.
- Gastrointestinal tract relies on sodium-glucose cotransporters (SGLTs)
- Cotransporters help move substances across membranes.

Indian Biological Data Center



- Indian Biological Data Center (IBDC) was inaugurated at Faridabad, Haryana
- First national repository for life science data
- Storage: Four-petabyte supercomputer '**Brahm**'
- Supported by: Department of Biotechnology (DBT).
- **Data sharing:** FAIR (Findable, Accessible, Interoperable and Reusable)
- **FAIR Data Principles:** support the reusability of digital assets.

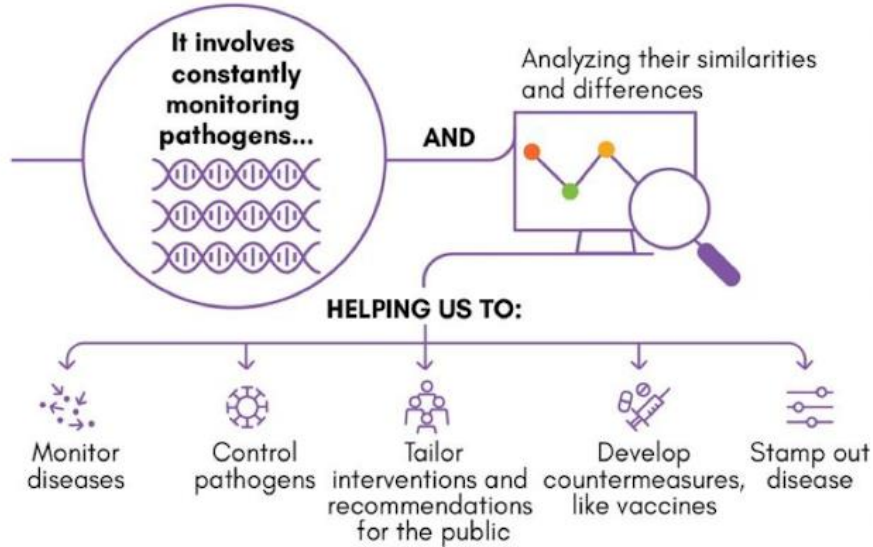
Indian Biological Data Center - Cont'd



- Also contains genomes sequenced by the Indian Sars-CoV-2 Genomic Consortium (INSACOG)
- Two sections have been developed:
 - Indian Nucleotide Data Archive (INDA) shall archive data generated from publicly funded research projects in India and provide internationally accepted data accession numbers
 - Indian Nucleotide Data Archive – Controlled Access (INDA-CA)

GENOMIC SURVEILLANCE

WHAT IS GENOMIC SURVEILLANCE?



APPLICATIONS OF GENOMICS:



Human genomics for medical purposes (Helping to treat genetic disorders)



Microbial genomics for medical purposes (identification of infectious agents, vaccine design etc.)



Agriculture and aquaculture genomics (identification of new traits and disease susceptibility etc.)



Biological and medical research (drug development, discovery of genes influencing disease etc.)



Others (forensic science, assessment of ancestry etc.)

GENOMIC SURVEILLANCE IN INDIA



Regional Genome Sequencing Laboratories (RGSL) serve as the regional hub for genome sequencing.



Indian SARS-CoV-2 Genomics Consortium (INSACOG) was established to expand Whole Genome Sequencing of SARS-CoV-2 across the nation*.



Indian Tuberculosis Genomic Surveillance Consortium (InTGS) is proposed on the lines of INSACOG.

*INSACOG, jointly initiated by Ministry of Health & Family Welfare, and Department of Biotechnology (DBT) with Council for Scientific & Industrial Research (CSIR) and Indian Council of Medical Research (ICMR), is a consortium of over 50 laboratories.

Dengue

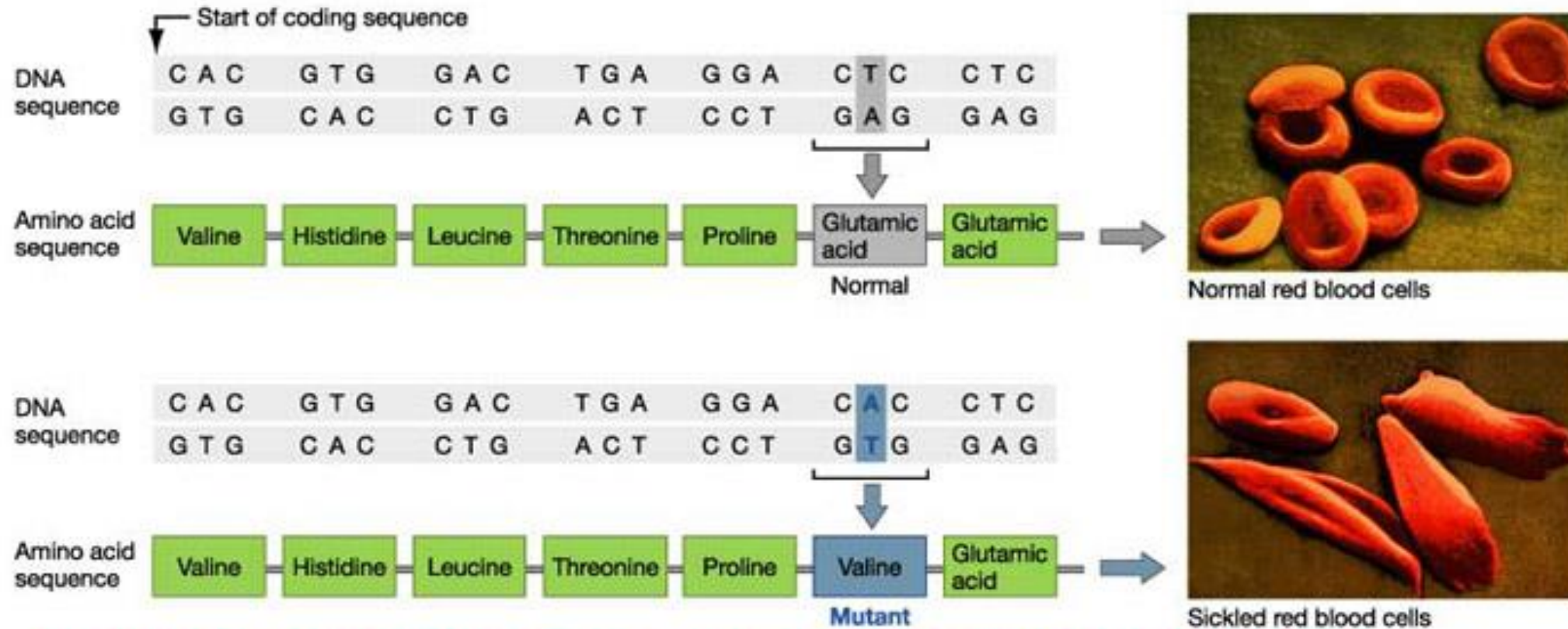


- Mosquito borne tropical disease caused by dengue virus (flavivirus)
- Spread through bite of infected aedes mosquito.
- First dengue vaccine : dengvaxia : approved by USFDA in 2019

Sickle Cell Anaemia



- Important this year : National Mission to Eliminate sickle cell anaemia launched in budget.
- Occurs due to genetic mutation that causes hemoglobin in RBCs to clump together.
- RBC shape changes to Sickle shape : flexibility lost.



The change in amino acid sequence causes hemoglobin molecules to crystallize when oxygen levels in the blood are low. As a result, red blood cells sickle and get stuck in small blood vessels.

Other Important News



- Trachoma : first country in southern Africa to eliminate trachoma is Malawi. India eliminated trachoma in 2017. Trachoma is an infection of the eye caused by Chlamydia Trachomatis. Can cause blindness.
- Diabetes : recent case of infection from diabetes medication. Diabetes is of 2 types : type 1 and type 2.
- Type 1 : insulin dependent, juvenile or childhood onset.
- Type 2 : Non insulin dependent : adult onset. (most common)
- Shigella : bacteria causing shigellosis. Leads to diarrhoea in children. No vaccine available.

Nanotechnology



- What we need to cover :
- Conceptual understanding of nanotech
- Current developments in nanotechnology



Nanotechnology : Basic Concepts



- Term Nanotech comes from the word nanometer
- Nanometer means 10^{-9}
- $10^{-1} = \frac{1}{10}$
- When we bring things to this size, it leads to certain special effects in them.

Special Effects in Nanomaterials



- Electrical Conductivity
- Color Effects
- Chemical Reactivity
- Melting Point
- Wettability

Special Effects : why and how

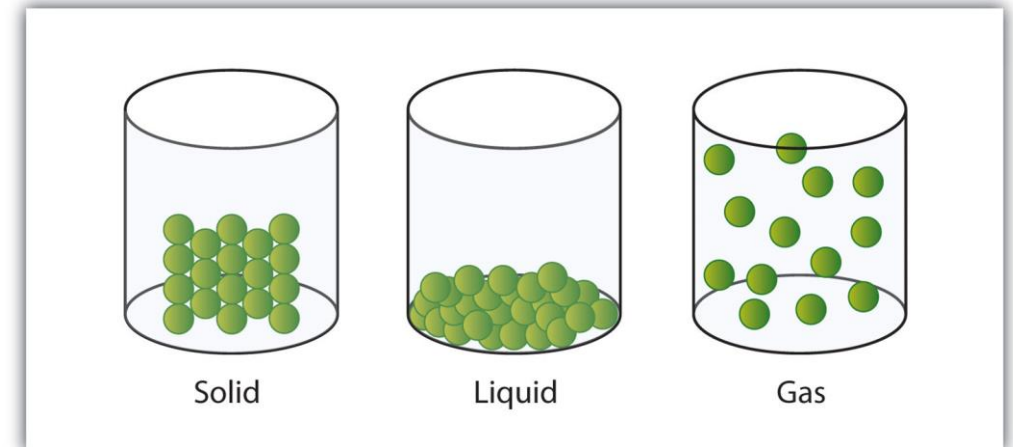


- Disclaimer : you don't need to know these concepts directly
- I am teaching this so that you can develop an understanding
- What do we already know?
- Atoms are constituent elements of materials
- Atoms have electrons, protons and neutrons.
- Solids liquids and gases have different arrangement of atoms.

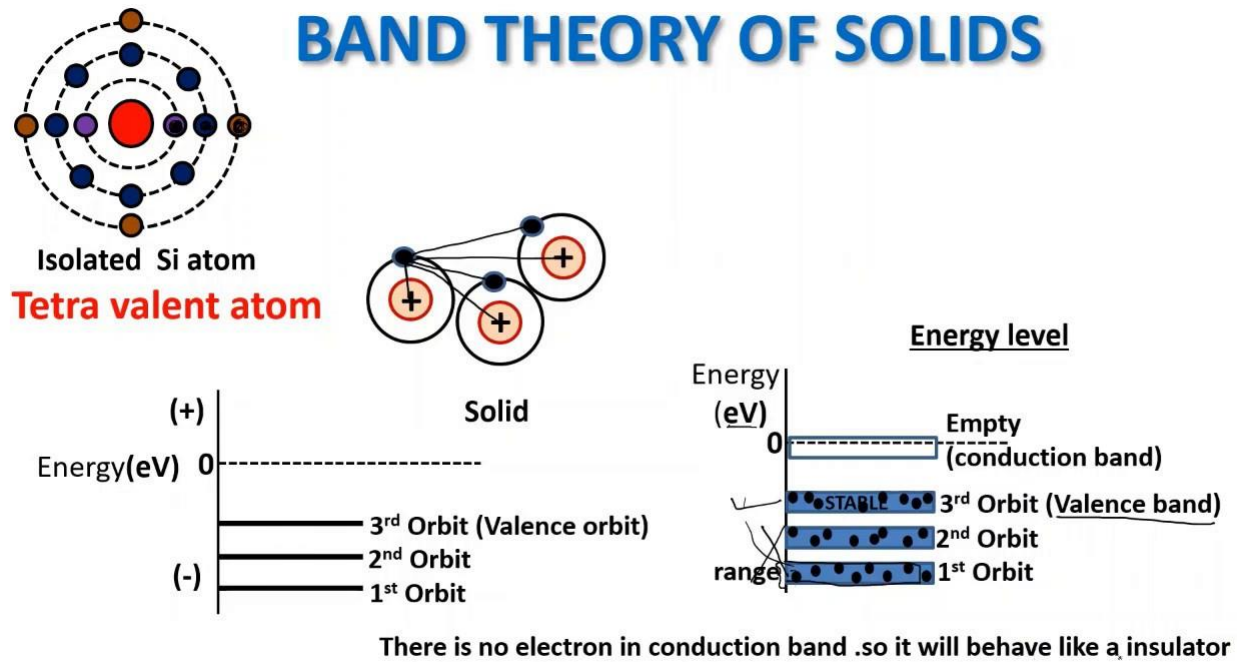
Concepts of Nanotech



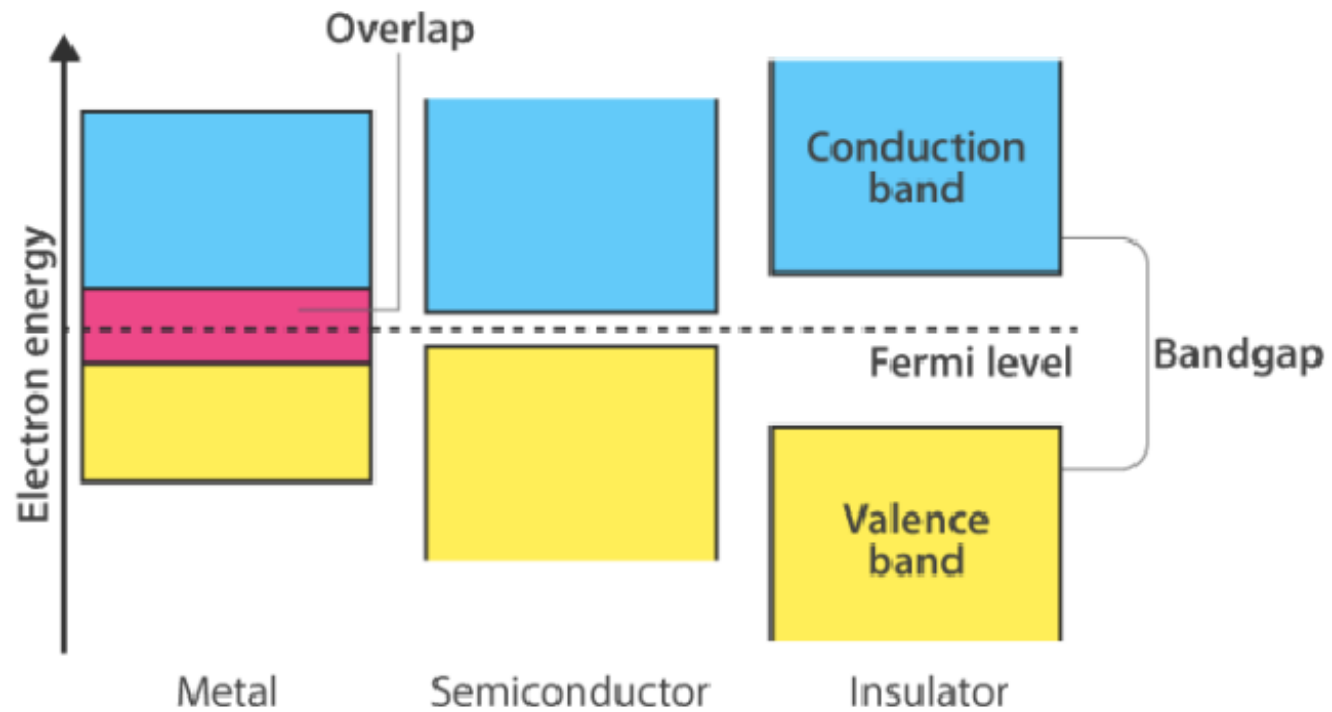
- We are concerned mainly with solids
- When atoms are packed together, their electrons come together into something known as bands
- This band arrangement is size dependent and it gets altered as size changes
- This is known as band theory of solids



Band theory of Solids



Bands

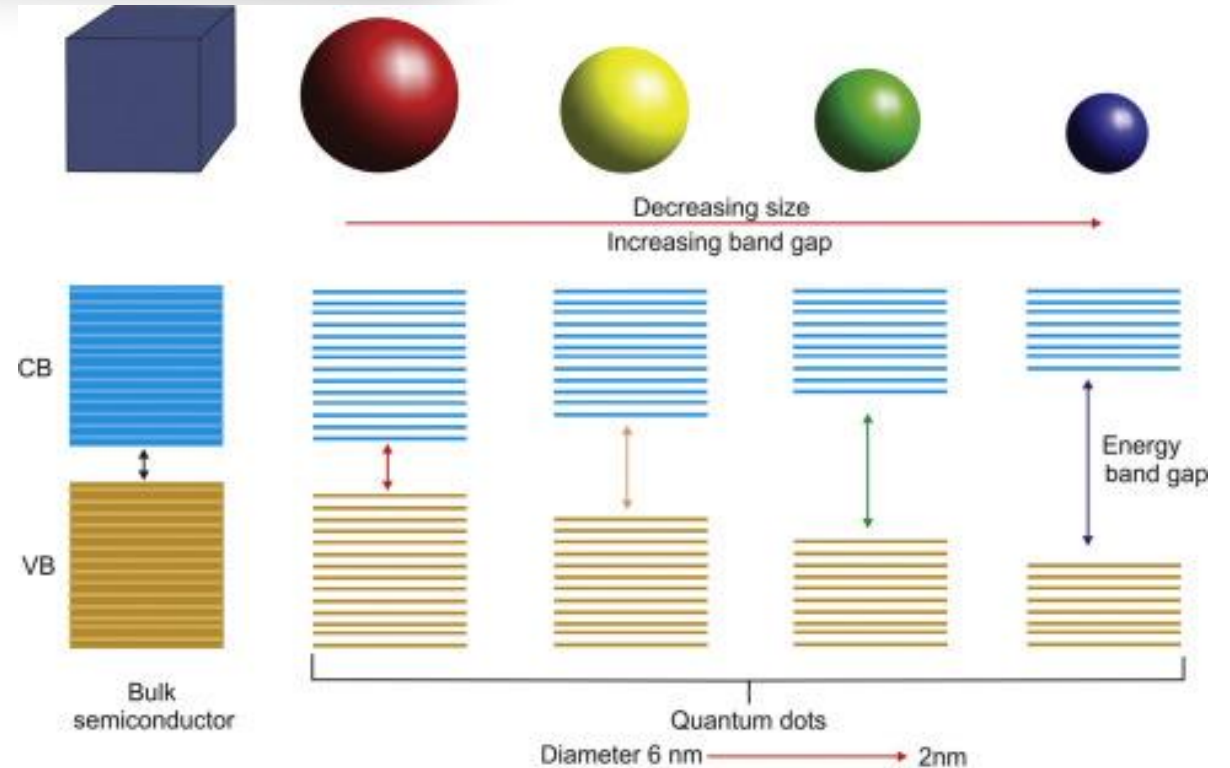


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Bands change at nano level



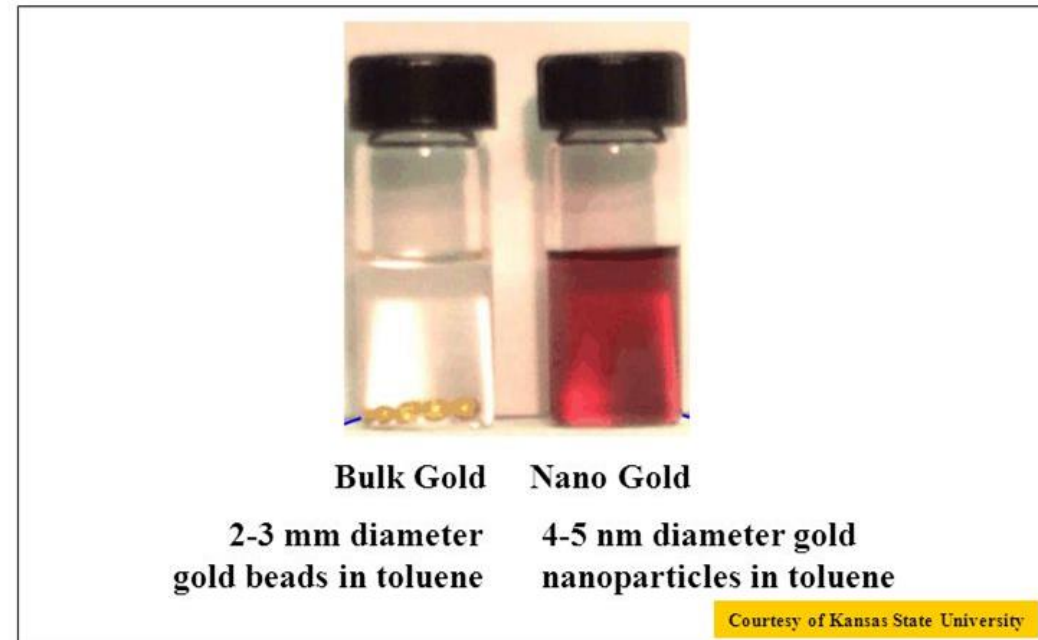
- Band arrangement changes at the level of nanomaterials
- As the bandgap increases, certain properties change
- Change in electrical conductivity : decrease in semi conductors
- Discrete energy levels lead to emission of pure wavelength of light : change in color



GOLD Particles



- Color is size dependent due to band gap alterations.



Extendable Structures: Melting Point, Color, Conductivity
© McREL 2009

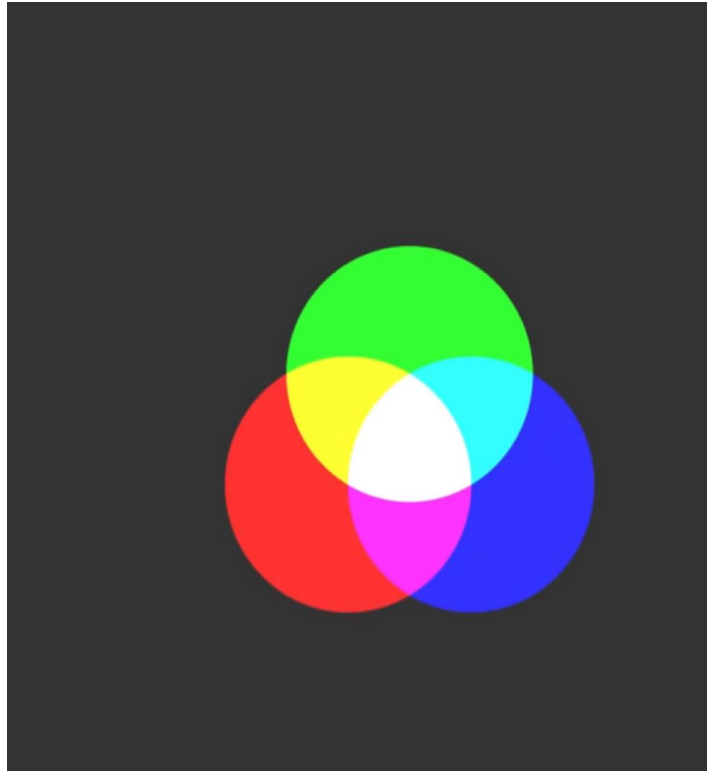
NanoLeap

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Change in optical properties



- From bands (combined) the energy states become discrete
- Hence the wavelengths emitted are pure
- Quantum Dots : This idea is used in the formation of Quantum LED (QLED)

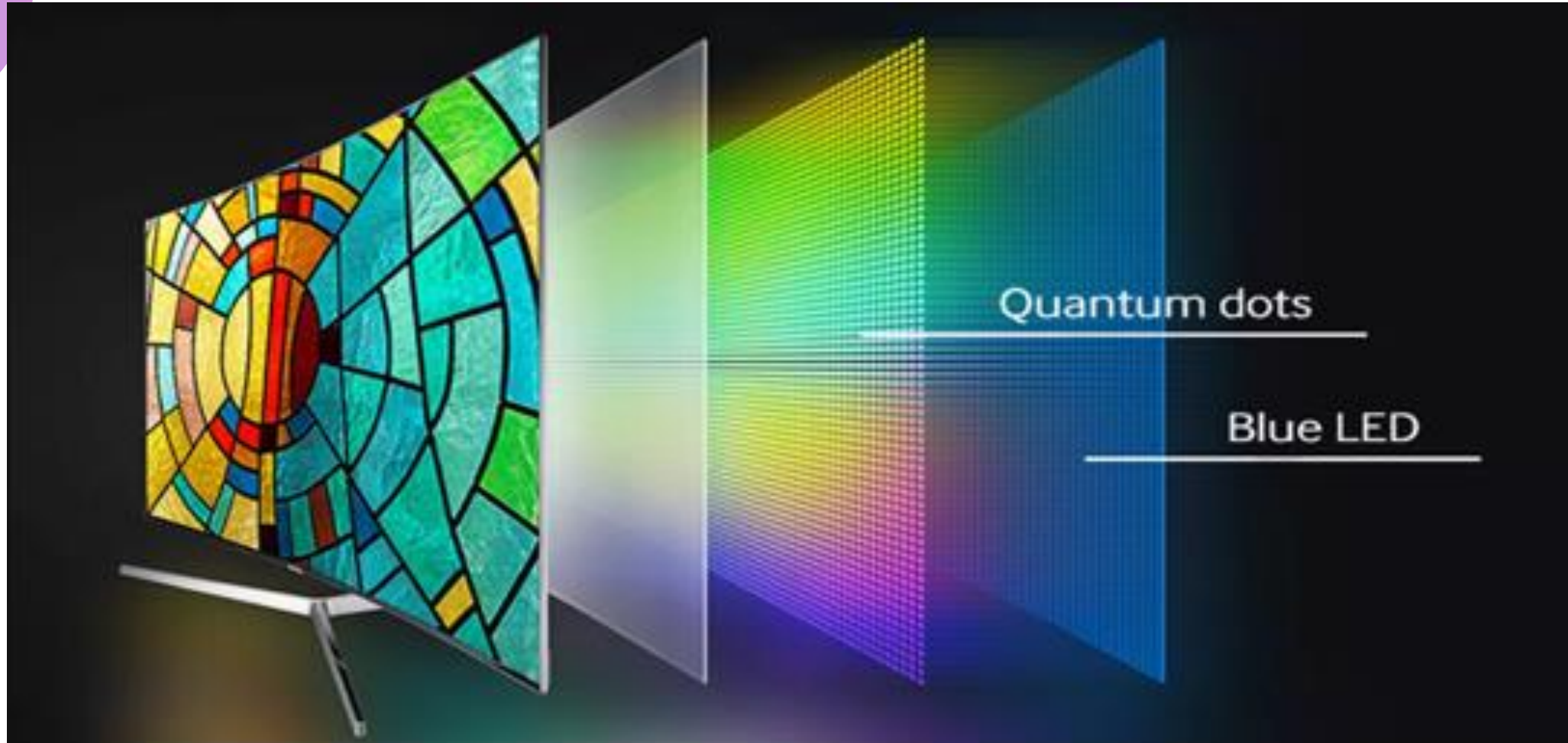


QLED

- Stands for Quantum dot Light Emitting Diode
- Uses quantum dot technology to emit light. Add an additional layer in front of a LED light panel
- Delivers 100% colour volume with unmatched saturation levels
- Not as power efficient as LED

LED

- Stands for Light Emitting Diode
- Uses LED to backlit the display
- Not as good as QLED in colour volume or picture quality
- Delivers optimal brightness level without consuming more power



Chemical Reactivity : change in nano materials



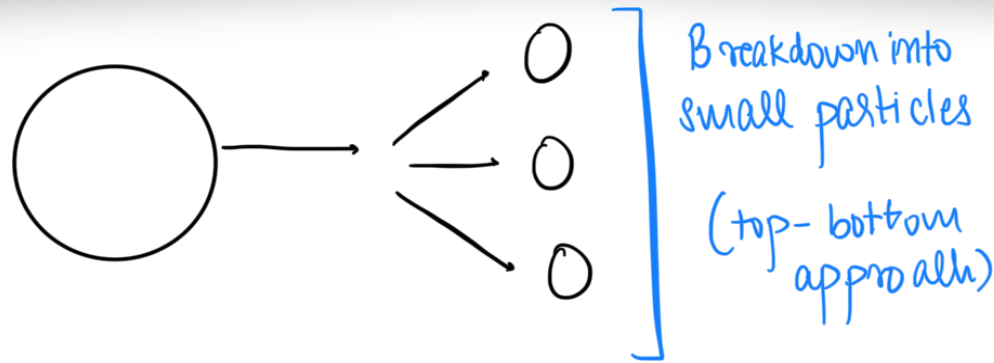
- Live demonstration : Disprin (solid vs crushed) in water
- More surface area : more reaction likelihood
- More reactivity
- Cube with side 4cm vs cube with side 1cm

Change in properties : why?

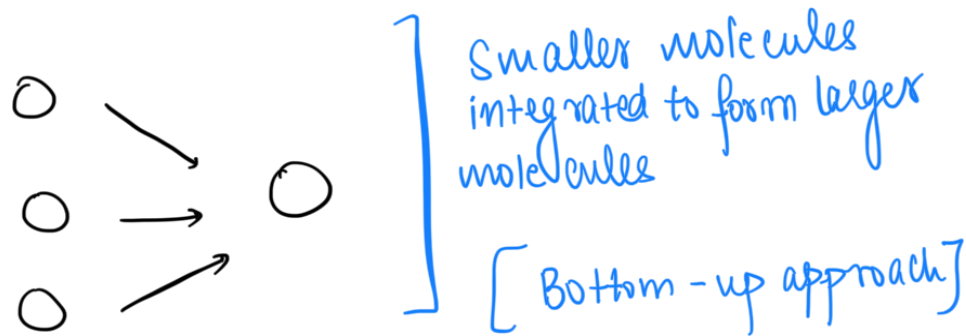


- These are known as quantum effects
- Quantum effects are size dependent
- Another concept : Quantum Confinement
- However, if you can just remember that there is a change in the properties of materials, it is sufficient.

2 approaches to make nanomaterials

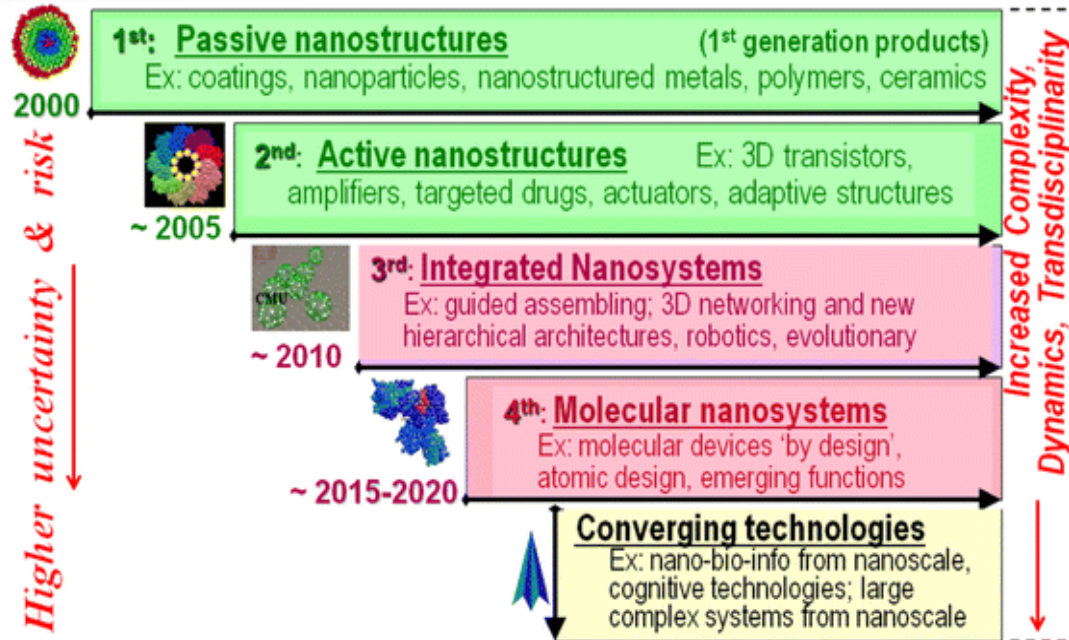


Example : nano catalyst,
nano silicon based solar
panels



Example :
Nanoelectronics and
integrated circuits

Generations of Nanomaterials



1. Passive Structures

1. Active : interact with environment and drug delivery

1. Nanosystems : Complete systems of nanomaterials (nobel prize of 2016)

1. Molecular nanosystems : bottom up approach and concept of additive manufacturing (3d printer)

2016 Chemistry Nobel Prize

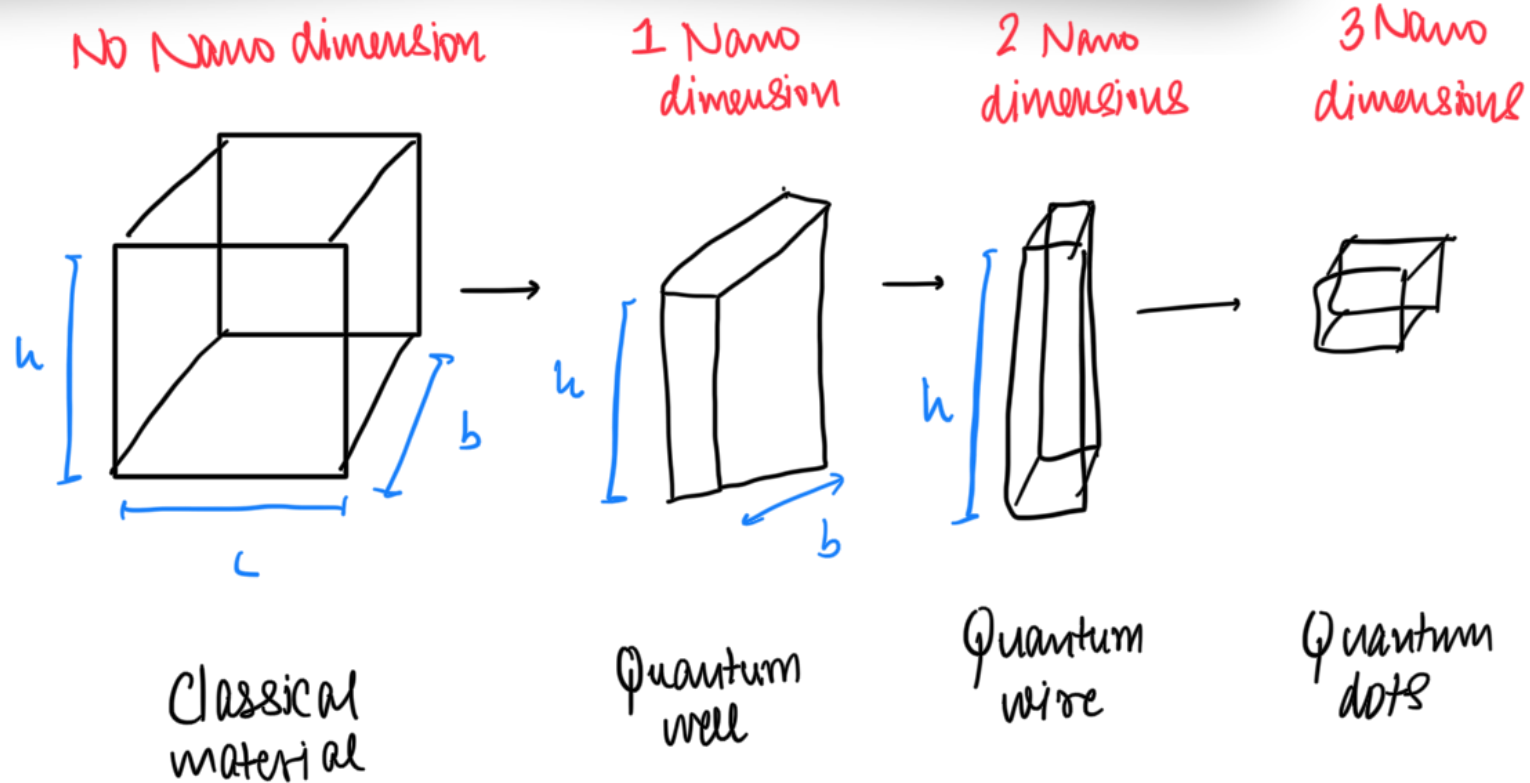


Nobel Prize in Chemistry / Winners (2016)

Fraser Stoddart, Jean-Pierre Sauvage, Ben Feringa

The **Nobel Prize in Chemistry 2016** is awarded to Jean-Pierre Sauvage, Sir J. Fraser Stoddart and Bernard L. Feringa for their design and production of molecular machines. They have developed molecules with controllable movements, which can perform a task when energy is added. Oct 5, 2016

Nanomaterials : Types



Applications of Nanotechnology



- This is where questions are mostly asked.
- Now that you have understood what nanotechnology is, understanding applications is easy.

NanoFertiliser



- Nano Urea Liquid : National Fertilisers Limited (NFL) and Rashtriya Chemicals and Fertilisers Ltd (RCF) signed MoU with IFFCO for 'transfer of technology' of Nano Urea Liquid (NUL) fertiliser
- NUL contains nano-scale nitrogen particles which have more surface area and number of particles making it more impactful.
- India has become the first country globally to start commercial production of NUL.

NanoFertiliser



Significance:

- Reduces the requirement of conventional Urea by 50% or more.
- Improves Soil, Air & Water quality.
- Cheaper than conventional urea. ***

Triboelectric NanoGenerators



- Scientists from the Centre for Nano and Soft Matter Sciences (CNSMS) have fabricated a cost-effective, bio-compatible nanogenerator that can generate electricity from mechanical energy in the form of vibrations present everywhere.
- CNSMS is an autonomous institute under the Department of Science & Technology.
- The energy harvesting TENG works on the principle of creation of electrostatic charges (deficiency or excess of electrons which occurs on ungrounded or insulating surfaces).
- Applications include optoelectronics, self-powered devices, and other biomedical applications.
- major application : pacemakers

Nanomicelles



- Recently, Researchers have found that Nanomicelles can be used for Cancer treatment.
- Nanomicelles are globe-like structures with a hydrophilic outer shell and a hydrophobic interior. This dual property makes them a perfect carrier for delivering drug molecules.
- Advantage: Low toxicity, ability to minimize drug degradation, ability to permeate tissues easily
- for drug delivery, and lower adverse drug side effects.
- Hydrophilic and hydrophobic configuration matches with phospholipid bilayer thereby allowing targeted drug delivery

Gold NanoParticles

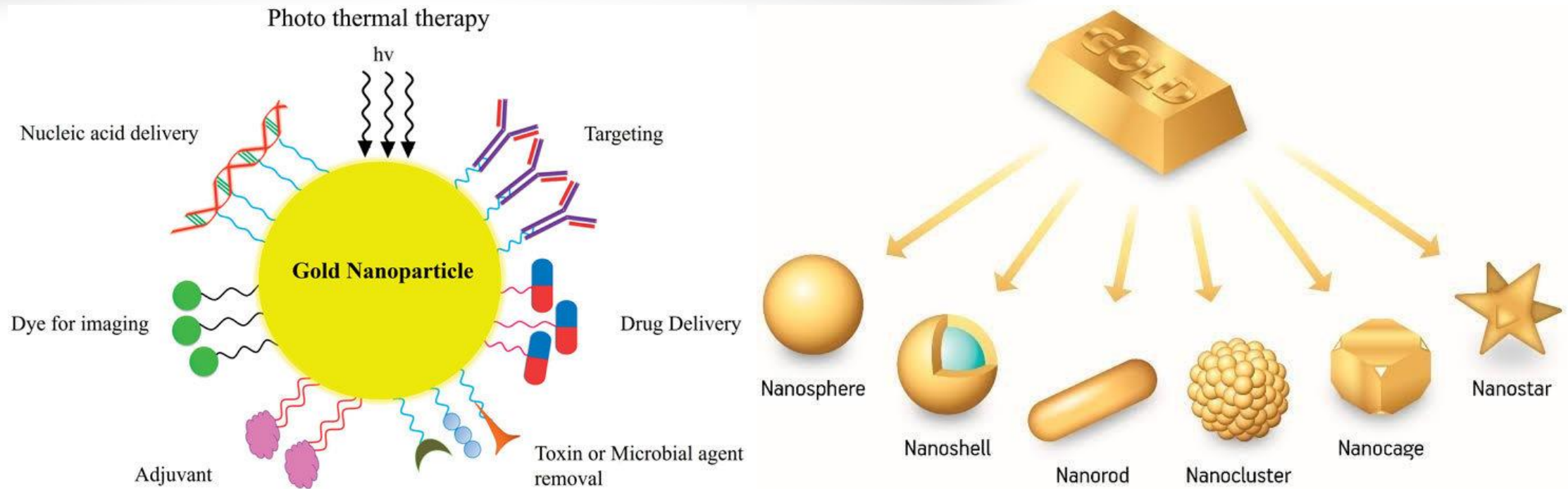


- National Centre for Polar and Ocean Research and the Goa University has successfully synthesized GNPs using psychrotolerant Antarctic bacteria.
- GNP's are biocompatible, have high surface area, more stability, and are non-toxic. GNPs are melted at much lower temperatures (300 °C) than bulk gold (1064 °C).
- Lets understand each of these words

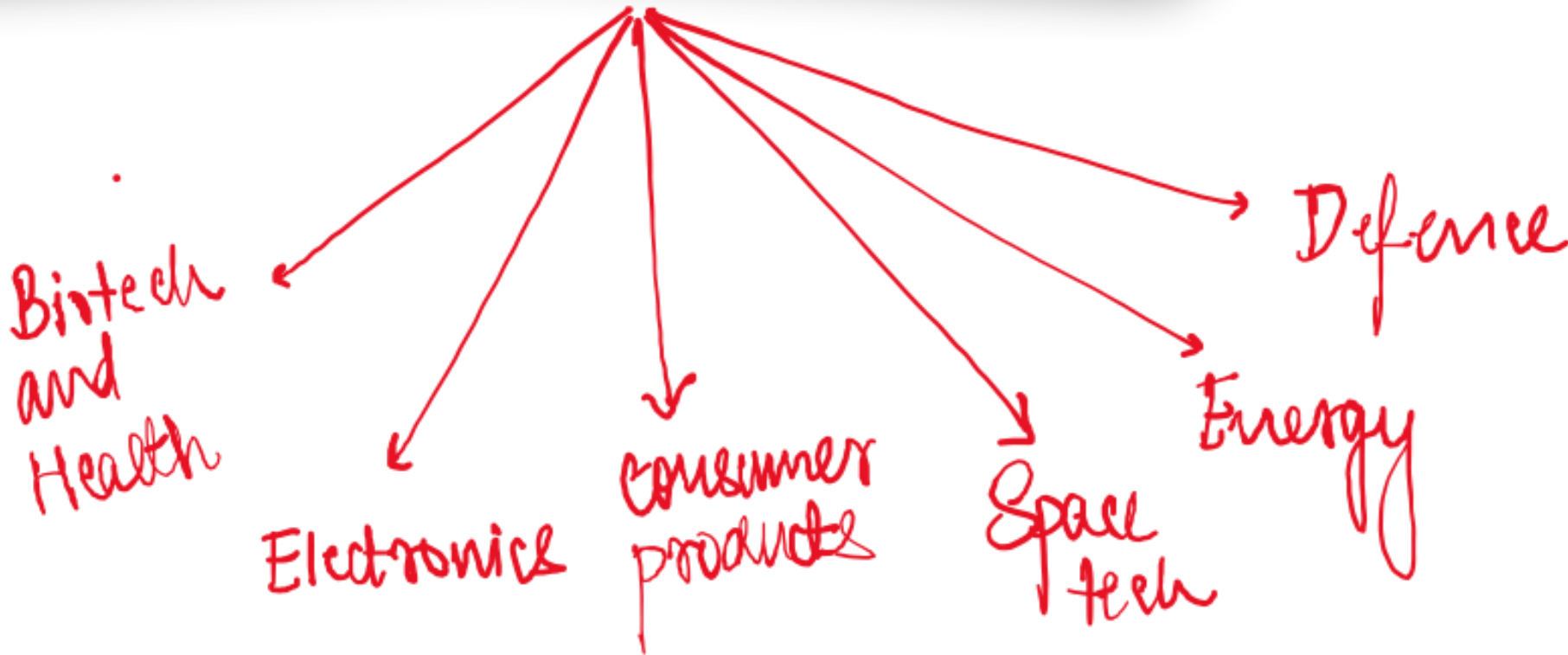
Advantages of GNP :

- Greater solar radiation absorbing ability : photovoltaic cells
- Unique optical properties : therapeutic imaging for better resolution
- Useful in the electronics industry : NOMFET (Nanoparticle Organic Memory Field-Effect Transistor) using GNPs. Mimics human synapse

Gold NanoParticles



Applications of Nanotech : enabling tech



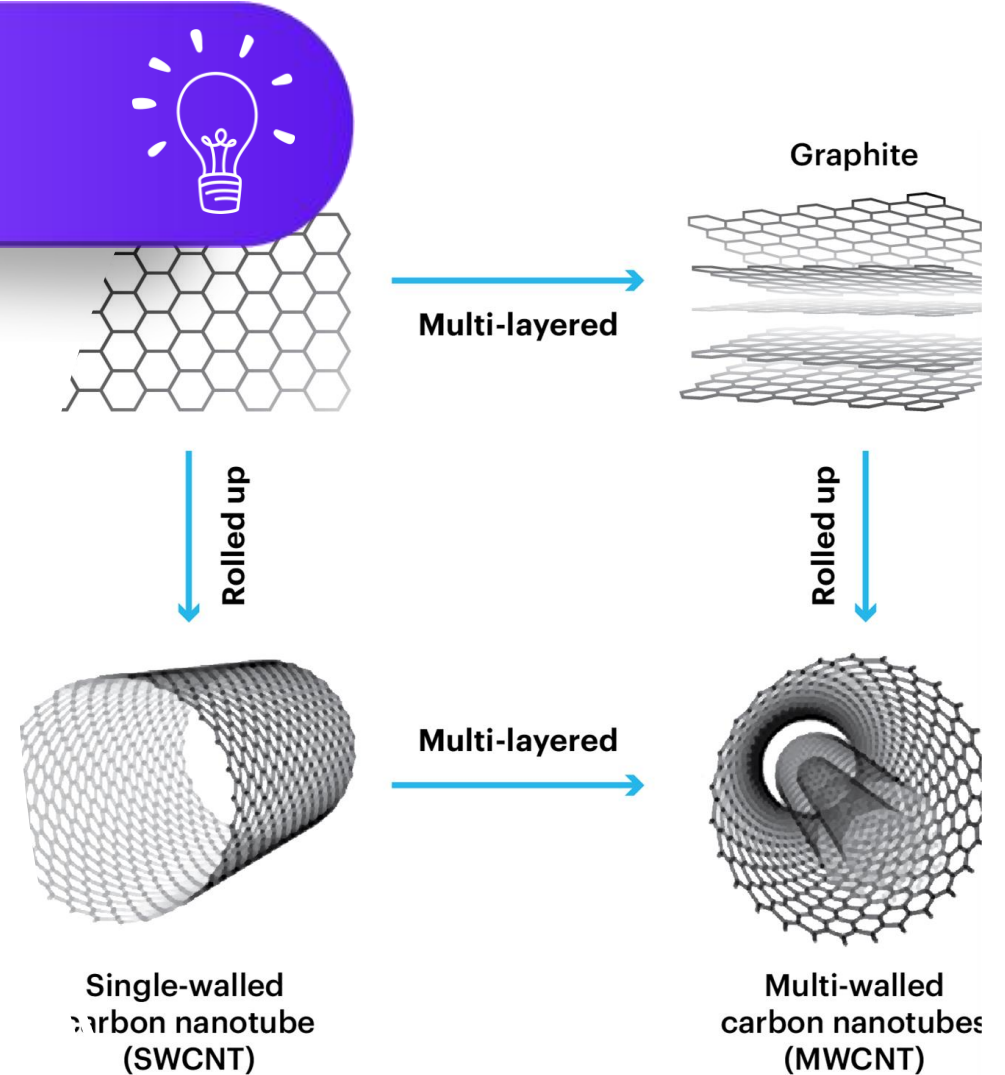
Carbon Nanomaterials



- C-60 Buckminster fullerene
- Carbon Nanotubes
- Graphene : 2d Graphite : Flexible touchscreens

Carbon Nanotubes

- Cylindrical Hollow figures with sheets of graphite
- Novel properties, lightweight
- Tensile strength : load bearing
- Electrical conductivity
- Used in electronics, automobiles, sports, space, aerospace industry
- However, not biocompatible : Immune Reactions



Applications : Industrial Products



- Sunscreens : Titanium dioxide and zinc oxide np help block uv radiations
- Non flammable furniture : coating
- Fabrics and Body armour : lightweight, wettability : easy clean textiles
- Films on glasses, computers and camera displays
- Quantum dots and electrical effects : QLED tv

Environmental Remediation



- Nanosensors : neutralize the effect of pollutants
- Wastewater treatments
- Nano membranes: desalinization

Energy Sector



- Biofuel production : nanobioengineering of enzymes
- Solar panels : TENG
- Hydrogen economy and storage systems
- Nanotube scrubbers and electrostatic precipitators for better efficiency
- GNP : solar panels
- CNT : PV cell
- Graphne : production of green hydrogen

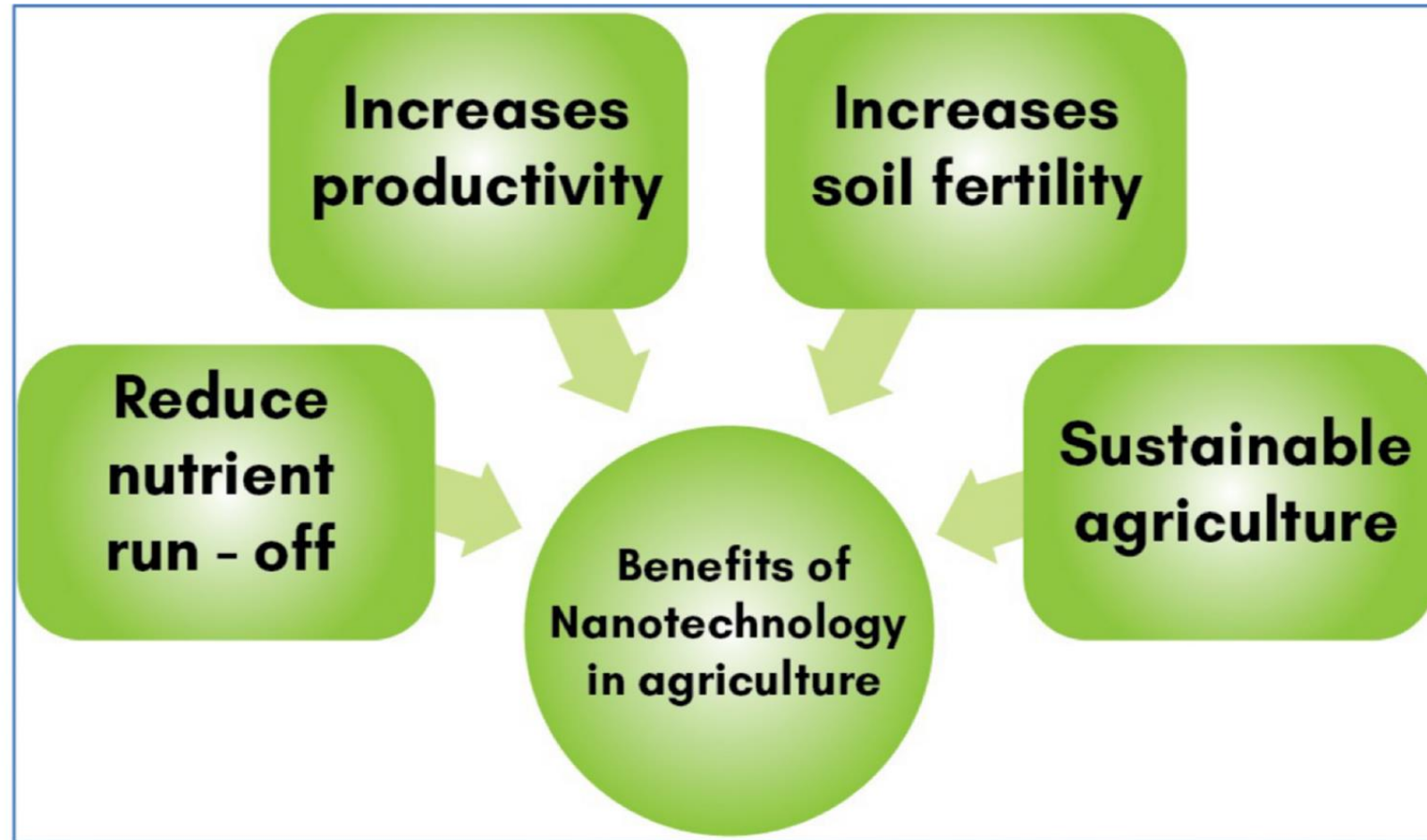
Agriculture Sector



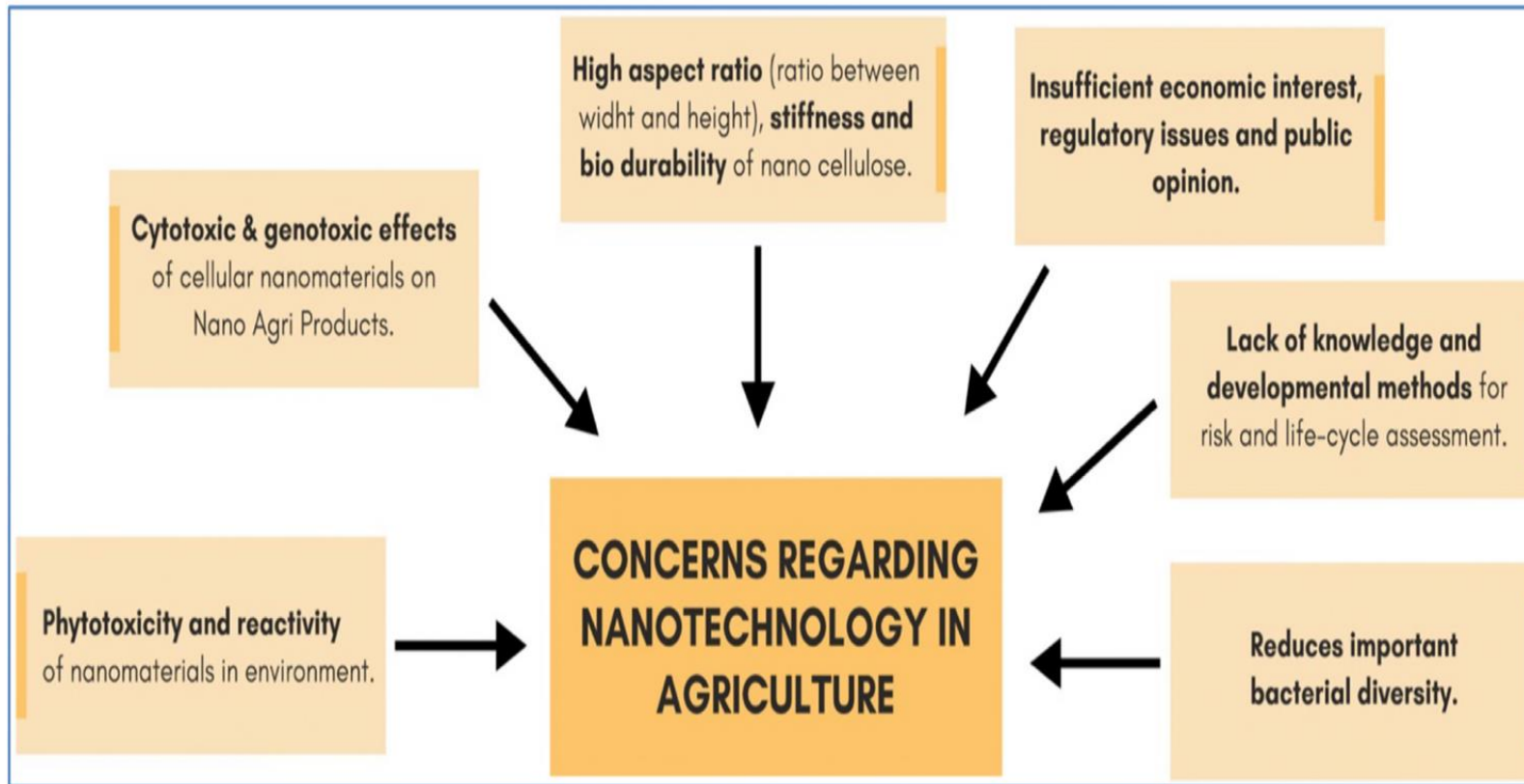
- Nanoherbicides and Nanopesticides : targeted delivery and enhancement of productivity
- Nanophosphorous fertilizer : Millet and cluster beans
- Silver Nanoparticles : against phytopathogens
- Carbon Nanoparticles : seed germination
- Nickel / ferrite : Antifungals
- Zinc oxide/ chitosan : Viral Diseases
- Precision farming : nanosensors

Nanotechnology in agriculture :

advantages :



Nanotechnology in agriculture : disadvantages :



Health Sector



- Antigen Antibody tests : ex : ELISA test (lets understand in detail)
- Prosthetic Devices : nanocomposite materials
- Targeted drug delivery : nanomicelles
- Nano sensors : HIV and Cancer detection
- GNP : medical imaging
- Nanohydrogel based eye drops

Defence



- Tank polymer and Camouflage
- Lightweight materials
- Precision guided missiles

Electronics



- Quantum dots
- Graphene : used in flexible touch screens
- Optoelectronics
- CNT : replace silicon based chips

Risks associated with nanotechnology



- Health Risks : generation of Free Radicals
- Biomagnification in the foodchain of nano particles
- Environmental Contamination
- High levels of complexity
- Lack of risk assessment methodologies
- Fears of self-replicating materials

India and Nanotechnology



- Nano Mission by Dr. CNR Rao in 2007
- The Government of India launched the Nano Mission in 2007 as an “umbrella capacity-building programme”.
- It is being implemented by the Department of Science and Technology (DST) under the Ministry of Science and Technology.
- The objectives of the mission are:
 - Basic research promotion
 - Infrastructure development
 - Nano applications and technology development

India and Nanotechnology



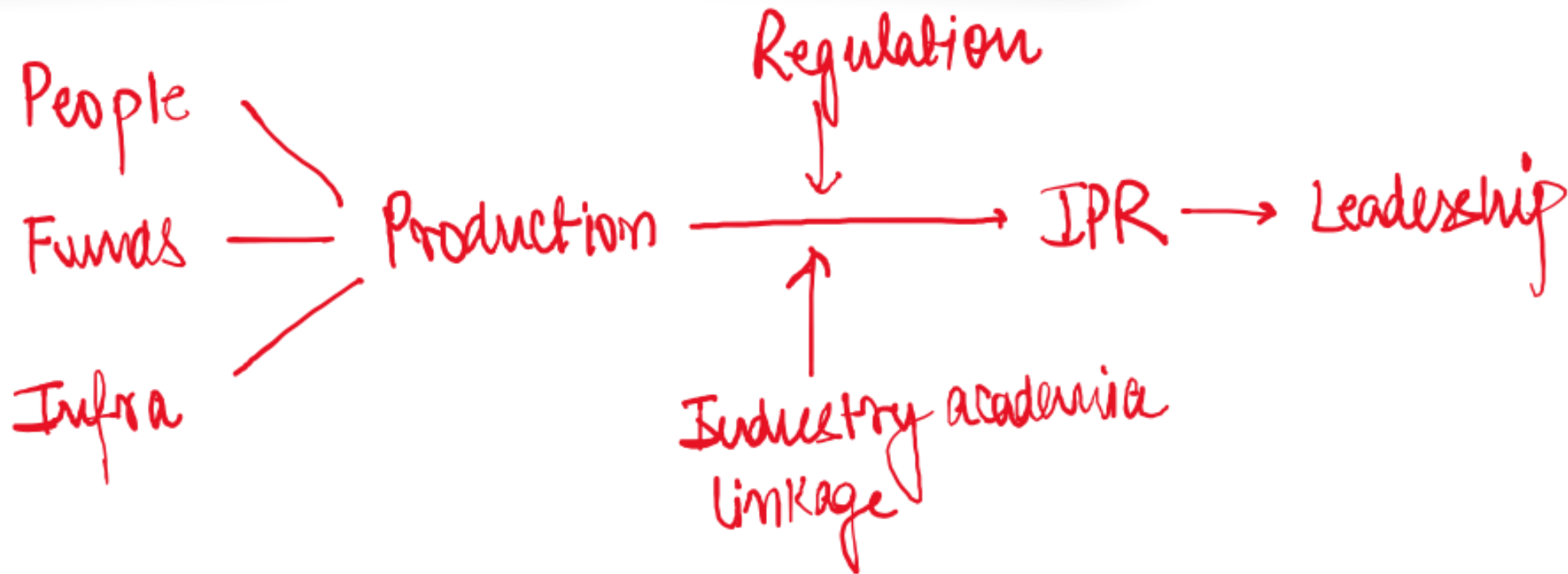
- Human Resource development
- International collaborations
- As a result of the efforts led by the Nano Mission, today, India is amongst the top five nations in the world in terms of scientific publications in nanoscience and technology (moving from 4th to the 3rd position).
- The Nano Mission has established national dialogues to promote R&D in the development of standards for nanotechnology and for laying down a National Regulatory Framework Road-Map for Nanotechnology (NRFR-Nanotech).

India and Nanotechnology



- ICONSAT : International Conference on NanoScience and NanoTechnology : biennial conferences under the aegis of nano mission.
- Center of Excellence in Nano Electronics : IIT Bombay and IISc bangalore

Way Forward: Life Cycle Approach



Test of 5 Questions!

Cut off : 5/5



Question 1 (2011)

Q : A company marketing food products advertises that its items do not contain trans-fats. What does this campaign signify to the customers?

1. food products are not made out of hydrogenated oils.
2. The food products are not made out of animal fats/ oils.
3. The oils used are not likely to damage the cardiovascular health of consumers.

Which of the statements given above is/are correct?

- (a) 1 only
- (b) 2 and 3 only
- (c) 1 and 3 only
- (d) 1, 2 and 3

Question 2 (2020)

- With reference to carbon nanotubes, consider the following statements
 1. They can be used as carriers of drugs and antigens in the human body.
 2. They can be made into artificial blood capillaries for an injured part of the human body.
 3. They can be used in biochemical sensors.
 4. Carbon nanotubes are biodegradable.

Q Which of the statements given above are correct?

- A) 1 and 2 only
- B) 2, 3 and 4 only
- C) 1, 3 and 4 only
- D) 1, 2, 3 and 4

Question 3 (2015)

- With reference to the use of nanotechnology in the health sector, which of the following statements is/are correct?

1. Targeted drug delivery is made possible by nanotechnology.
2. Nanotechnology can largely contribute to gene therapy.

Q : Select the correct answer using the code given below.

- A) 1 only
- B) 2 only
- C) Both 1 and 2
- D) Neither 1 nor 2

Question 4 (2012)

- Graphene is frequently in news recently. What is its importance?
 1. It is a two-dimensional material and has good electrical conductivity.
 2. It is one of the thinnest but strongest materials tested so far.
 3. It is entirely made of silicon and has high optical transparency.
 4. It can be used as 'conducting electrodes' required for touch screens, LCDs and organic LEDs.

Which of the statements given above are correct?

- a) 1 and 2 only
- b) 3 and 4 only
- c) 1, 2 and 4 only
- d) All of the above

Question 5 (2022)

Consider the following statements:

1. Other than those made by humans, nanoparticles do not exist in nature.
2. Nanoparticles of some metallic oxides are used in the manufacture of some cosmetics.
3. Nanoparticles of some commercial products which enter the environment are unsafe for humans.

Which of the statements given above is/are correct?

- (a) 1 only
- (b) 3 only
- (c) 1 and 2
- (d) 2 and 3

Congratulations on giving the test!

Let's Discuss The Answers.



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